SURJECTIVITY OF THE MAP ON SPECTRA IS EQUIVALENT TO THE CONTRACTION OF EXTENDED IDEALS BEING THE IDEALS THEMSELVES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof refers to two same-chapter lemmas by label; both are recorded as navigation stubs pointed at their \label lines, and the arguments consuming them stay inside the counted span.

Editorial note. The counted spans contain no \cite, so the card carries no bibliography.

Allowed standard prerequisites (editorial).

- Radical of an ideal and its behaviour under inverse image. Used at the start of the counted proof (line 6057). Shows the conditions on the radicals of extended ideals and on radical ideals are equivalent.
- Membership in the image via nonvanishing after tensoring with the residue field. Invoked in the counted proof at lines 6064-6067 and again at the end. Connects the contraction condition for primes with the topological surjectivity statement.

Source reference. Original lemma 00E0: sources/stacks-algebra/pinned-original-source.tex, source lines 3204–3204. Navigation pointer for \ref{lemma-Zariski-topology} used inside the counted span (line 6062); the argument that consumes it is retained in the span.

Source reference. Original lemma 00E7: sources/stacks-algebra/pinned-original-source.tex, source lines 3717–3717. Navigation pointer for \ref{lemma-in-image}

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used inside the counted span (line 6065); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $\varphi : R \to S$ be a ring map. The following are equivalent:*

1. *The map $\mathrm{Spec}(S) \to \mathrm{Spec}(R)$ is surjective.*
2. *For any ideal $I \subset R$ the inverse image of $\sqrt{IS}$ in R is equal to $\sqrt{I}$.*
3. *For any radical ideal $I \subset R$ the inverse image of IS in R is equal to I.*
4. *For every prime $\mathfrak{p}$ of R the inverse image of $\mathfrak{p}S$ in R is $\mathfrak{p}$.*

In this case the same is true after any base change: Given a ring map $R \to R'$ the ring map $R' \to R' \otimes_R S$ has the equivalent properties (1), (2), (3) as well.

Proof. If $J \subset S$ is an ideal, then $\sqrt{\varphi^{-1}(J)} = \varphi^{-1}(\sqrt{J})$. This shows that (2) and (3) are equivalent. The implication (3) $\Rightarrow$ (4) is immediate. If $I \subset R$ is a radical ideal, then Lemma 00E0 guarantees that $I = \bigcap_{I \subset \mathfrak{p}} \mathfrak{p}$. Hence (4) $\Rightarrow$ (2). By Lemma 00E7 we have $\mathfrak{p} = \varphi^{-1}(\mathfrak{p}S)$ if and only if $\mathfrak{p}$ is in the image. Hence (1) $\Leftrightarrow$ (4). Thus (1), (2), (3), and (4) are equivalent.

Assume (1) holds. Let $R \to R'$ be a ring map. Let $\mathfrak{p}' \subset R'$ be a prime ideal lying over the prime $\mathfrak{p}$ of R. To see that $\mathfrak{p}'$ is in the image of $\mathrm{Spec}(R' \otimes_R S) \to \mathrm{Spec}(R')$ we have to show that $(R' \otimes_R S) \otimes_{R'} \kappa(\mathfrak{p}')$ is not zero, see Lemma 00E7. But we have

$$(R' \otimes_R S) \otimes_{R'} \kappa(\mathfrak{p}') = S \otimes_R \kappa(\mathfrak{p}) \otimes_{\kappa(\mathfrak{p})} \kappa(\mathfrak{p}')$$

which is not zero as $S \otimes_R \kappa(\mathfrak{p})$ is not zero by assumption and $\kappa(\mathfrak{p}) \to \kappa(\mathfrak{p}')$ is an extension of fields. □